A REFUTATION OF THE PROOF IN "ON ARITHMETIC PROGRESSIONS AND A PROOF OF THE NONEXISTENCE OF MAGIC SQUARES OF SQUARES" (ARXIV:2510.08286)

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ABSTRACT. The preprint arXiv:2510.08286 (v3, April 2026) claims a proof that no 3×3 magic square of distinct square integers exists. We show that the proof is invalid. Its penultimate equation, eq. (29), is proven here to equal a strictly positive cofactor times the preprint's own Lemma-3.2 spacing constraint — an algebraic identity valid over any ordered field, carrying no Diophantine information (Theorem A) — and the final inference drawn from (29), namely $\beta_{1d}^2 - \beta_{1n}^2 = 0$ by comparison of coefficients of odd powers of α_{1d}, fails on an explicit configuration satisfying every hypothesis and every positivity condition that the inference invokes (Theorem B). The witness is itself of independent interest: a magic square with six perfect-square entries produced by the preprint's own (correct) framework. We emphasize that this note refutes the *proof*, not the *statement*: the existence question for 3×3 magic squares of squares remains open. All computations are exact and are reproduced by a short script given in Appendix A.

1. Introduction

A 3×3 *magic square of squares* is a 3×3 array of nine distinct perfect squares whose rows, columns and both diagonals share a common sum. Whether one exists over the integers is a well-known open problem (LaBar [2]; Guy [3], Problem D15). The preprint [1] (henceforth "the preprint"; we refer throughout to version 3, dated April 7, 2026) claims a proof of nonexistence as its Theorem 3.1.

This note establishes:

(a) the preprint's reduction framework is *correct and complete* — its constraint system is equivalent to the classical Lucas structure of the problem (Proposition 1), and its algebra is correct through its eq. (30) inclusive;

(b) its eq. (29) is a positive multiple of its own Lemma-3.2 constraint, hence equivalent to it, hence devoid of arithmetic content beyond it (Theorem A);

(c) the final inference — from (29) to $\beta_{1d}^2 - \beta_{1n}^2 = 0$ — is invalid: an explicit configuration satisfies all of its hypotheses and positivity

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conditions while violating its conclusion, with both sides of (29) nonzero (Theorem B).

Consequently the proof of Theorem 3.1 of [1] does not go through, and, as we argue in §6, no local repair is possible: the derivation never leaves the real-algebraic world, where the constraint system is abundantly solvable.

Throughout, equation numbers (2), (6)–(14), (16)–(30) refer to the preprint's numbering; our own statements are labelled Proposition 1, Theorem A, Theorem B.

2. The preprint's framework, recalled

For reals $0 < p < q < r$ with

$$q^2 - p^2 \;=\; r^2 - q^2 \;=\; \Sigma\mathcal{P},$$

the preprint calls the corresponding pair of consecutive runs of odd numbers an *AP pair* $\mathcal{P}$ with sum $\Sigma\mathcal{P}$, and associates to it (its §2):

- lengths $\mathcal{P}^{n_1} = q - p$ and $\mathcal{P}^{n_2} = r - q$, with $\kappa = \mathcal{P}^{n_1}/\mathcal{P}^{n_2} > 1$;
- $\alpha = \kappa(\kappa+1)/(\kappa-1)$, written α_n/α_d (its eq. (9)); offsets $\mathcal{P}^1 = 2p$, $\mathcal{P}^2 = 2q$, $\mathcal{P}^3 = 2r - 2$;
- $s = \sqrt{(\alpha_n - 3\alpha_d)^2 - 8\alpha_d^2}$ and N satisfying $N(N+2s) = 8\alpha_d^2$, i.e. a root of $N^2 - 2(\alpha_n - 3\alpha_d)N + 8\alpha_d^2 = 0$ (its eqs. (10)–(12); both roots satisfy (12), and (13), (14) both recover κ).

Three consequences of these definitions, each a short computation and each verified mechanically at every configuration appearing below:

(∗)

$$\Sigma\mathcal{P} = \alpha\,(\mathcal{P}^{n_2})^2, \qquad q = \tfrac{1}{2}\,\mathcal{P}^{n_2}(\alpha - 1), \qquad p = \frac{\mathcal{P}^{n_2}\,s}{2\alpha_d}, \qquad \text{with } s = \frac{8\alpha_d^2 - N^2}{2N}.$$

(The first two are the preprint's (6) and (8); the third is equivalent to its (7).)

For a hypothetical 3×3 magic square of squares the preprint (its §3) derives three *distinct* AP pairs $\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3$ with equal sums D_1 (its (17)–(20)), sets

$$\beta_1 = \frac{\mathcal{P}_1^{n_2}}{\mathcal{P}_2^{n_2}} = \sqrt{\frac{\alpha_2}{\alpha_1}}, \qquad \beta_2 = \frac{\mathcal{P}_2^{n_2}}{\mathcal{P}_3^{n_2}} = \sqrt{\frac{\alpha_3}{\alpha_2}}$$

(its (21), (22)), splits them as

$$\beta_{1n} = \sqrt{\alpha_{2n}/\alpha_{1n}}, \quad \beta_{1d} = \sqrt{\alpha_{2d}/\alpha_{1d}}, \quad \beta_{2n} = \sqrt{\alpha_{3n}/\alpha_{2n}}, \quad \beta_{2d} = \sqrt{\alpha_{3d}/\alpha_{2d}}$$

— "where $\beta_{1n}, \beta_{1d}, \beta_{2n}, \beta_{2d} \in \mathbb{R}^+$" (its (23)–(26), p. 5) — and proves (its Lemma 3.2, whose case analysis is correct) the spacing constraint

$$\text{(L3.2)} \quad \Sigma\mathbf{p}_A = \Sigma\mathbf{p}_B, \qquad \text{equivalently} \qquad E := (p_2^2 - q_1^2) - (p_3^2 - q_2^2) = 0.$$

From these ingredients it asserts its eq. (29):

$$\text{(29)} \quad N_3^2\beta_{2n}^2\beta_{2d}^2\big(8\beta_{1d}^4\alpha_{1d}^2 - N_2^2\big)^2 - N_2^2\big(8\beta_{2d}^4\beta_{1d}^4\alpha_{1d}^2 - N_3^2\big)^2$$
$$= 4N_2^2N_3^2\beta_{1n}^2\beta_{1d}^2\beta_{2n}^2\beta_{2d}^2(\alpha_{1n} - \alpha_{1d})^2 - 4N_2^2N_3^2\beta_{2n}^2\beta_{2d}^2\big(\beta_{1n}^2\alpha_{1n} - \beta_{1d}^2\alpha_{1d}\big)^2,$$

and concludes (p. 7; quoted in full in §6) that since the left-hand side contains only even powers of α_{1d}, the coefficients of odd powers of α_{1d} on the right-hand side must vanish, forcing $\beta_{1d}^2 - \beta_{1n}^2 = 0$, hence $\beta_1 = 1$, hence $\mathcal{P}_1 = \mathcal{P}_2 = \mathcal{P}_3$, contradicting distinctness.

3. What is correct in the preprint

Fairness requires stating what checks out, and it is most of the preprint.

Proposition 1. *The constraint system of* [1] *— three AP pairs with equal sums together with the spacing constraint* (L3.2) *— is equivalent to the following: the nine values form a 3×3 additive grid $M + iD + jF$, $(i, j) \in \{0, 1, 2\}^2$. This is precisely the classical Lucas structure of a 3×3 magic square, the grid being arranged into the square by a pair of orthogonal Latin squares.*

Proof sketch. Given the chain of the preprint's (16), write the nine values as three triples (a_j, b_j, c_j) with $b_j - a_j = c_j - b_j = D$ $(j = 1, 2, 3)$ and $a_2 - b_1 = a_3 - b_2 = E$; direct chaining gives value$(i, j) = a_1 + iD + j(D + E)$, an additive grid with $F := D + E$. Conversely the grid values $c + x + y$, $x \in \{-D, 0, D\}$, $y \in \{-F, 0, F\}$, arrange into the magic square with center c in the standard way, every line summing to $3c$. □

Two consequences. First, the reduction of the preprint's §3 is faithful *and complete*: nothing is lost. Second — a point in the preprint's favor — *no integer counterexample to its hypotheses can exist*, since nine distinct integer squares satisfying them would constitute a magic square of squares; any flaw in the proof must therefore be inferential, which is what we exhibit.

Moreover, the preprint's algebra is correct through its eq. (30) inclusive: its Lemma 3.2 and case analysis are sound; eq. (29) is a genuine consequence of the framework (Theorem A below states precisely which consequence); and the printed (30) is the correct expansion of the right-hand side of (29), which factors as

$$\text{RHS}(29) \;=\; 4N_2^2N_3^2\,\beta_{2n}^2\beta_{2d}^2\left(\beta_{1d}^2 - \beta_{1n}^2\right)\left(\beta_{1n}^2\alpha_{1n}^2 - \beta_{1d}^2\alpha_{1d}^2\right), \tag{1}$$

the bracket of (30) being the second factor expanded via its eq. (12). The error is confined to the single inference drawn after (30).

4. Equation (29) is the constraint (L3.2) in costume

Since every β enters (29) squared, write

$$b_{1n} = \beta_{1n}^2, \quad b_{1d} = \beta_{1d}^2, \quad b_{2n} = \beta_{2n}^2, \quad b_{2d} = \beta_{2d}^2, \qquad n = \alpha_{1n}, \quad d = \alpha_{1d},$$

so that $\alpha_{2d} = b_{1d}d$, $\alpha_{3d} = b_{2d}b_{1d}d$, $\alpha_{2n} = b_{1n}n$.

Theorem A. *As polynomials in $\mathbb{Z}[n, d, N_2, N_3, b_{1n}, b_{1d}, b_{2n}, b_{2d}]$,*

$$\text{LHS}(29) - \text{RHS}(29) \;=\; K_0 \cdot \frac{4E}{t^2}, \qquad K_0 := 4N_2^2N_3^2\,b_{1d}^2b_{2d}^2\,d^2 = 4N_2^2N_3^2\,\beta_{1d}^4\beta_{2d}^4\,\alpha_{1d}^2,$$

where $t = \mathcal{P}_3^{n_2}$ is the free scale and $4E/t^2$ is the constraint (L3.2) *expressed in the same variables via* ($*$) *and the preprint's (21)–(26), with denominators cleared.*

Proof. By ($*$) and the β-substitutions, with $t := \mathcal{P}_3^{n_2}$, $\mathcal{P}_2^{n_2} = \beta_2 t$, $\mathcal{P}_1^{n_2} = \beta_1\beta_2 t$:

$$\frac{4E}{t^2} = T_1 - T_2 - T_3 + T_4,$$

where

$$T_1 = \frac{4p_2^2}{t^2} = \frac{b_{2n}}{b_{2d}} \cdot \frac{(8b_{1d}^2 d^2 - N_2^2)^2}{4N_2^2\, b_{1d}^2 d^2}, \qquad T_2 = \frac{4q_1^2}{t^2} = \frac{b_{1n}b_{2n}}{b_{1d}b_{2d}} \cdot \frac{(n-d)^2}{d^2},$$

$$T_3 = \frac{4p_3^2}{t^2} = \frac{(8b_{2d}^2 b_{1d}^2 d^2 - N_3^2)^2}{4N_3^2\, b_{2d}^2 b_{1d}^2 d^2}, \qquad T_4 = \frac{4q_2^2}{t^2} = \frac{b_{2n}}{b_{2d}} \cdot \frac{(b_{1n}n - b_{1d}d)^2}{b_{1d}^2 d^2}.$$

Multiplying each term by K_0 gives, term for term,

$$K_0T_1 = N_3^2\, b_{2n}b_{2d}\,(8b_{1d}^2 d^2 - N_2^2)^2, \qquad K_0T_3 = N_2^2\,(8b_{2d}^2 b_{1d}^2 d^2 - N_3^2)^2,$$

$$K_0T_2 = 4N_2^2N_3^2\, b_{1n}b_{1d}b_{2n}b_{2d}\,(n-d)^2, \qquad K_0T_4 = 4N_2^2N_3^2\, b_{2n}b_{2d}\,(b_{1n}n - b_{1d}d)^2,$$

which are exactly the four terms of (29) with matching signs: $K_0(T_1 - T_3) =$ LHS(29) and $K_0(T_2 - T_4) =$ RHS(29). ☐

Corollary A.1. *Since $K_0 > 0$ on every configuration:*

1. *eq. (29) is* equivalent *to the constraint* (L3.2)*. It holds on every configuration satisfying the framework of §2 — over any ordered field, in particular over $\mathbb{R}$ — and fails the moment* (L3.2) *is broken. It carries no arithmetic information beyond* (L3.2).
2. *The entire derivation of the preprint's §§2–3, through its eq. (30), consists of identities valid over $\mathbb{R}$. Over $\mathbb{R}$ the hypothesis system is abundantly solvable (by Proposition 1, any nine distinct positive reals in the grid pattern are squares of reals), with $\beta_1 \neq 1$ generically. Hence no inference valid over $\mathbb{R}$ can complete the proof from (29); a correct completion must use integrality essentially.*

5. The counterexample to the final inference

Theorem B. *There is a configuration satisfying every hypothesis and every positivity condition invoked by the final step of* [1] *on which eq. (29) holds with both sides nonzero and $\beta_1 = 6/5 \neq 1$. Consequently the inference "(29) $\Rightarrow \beta_{1d}^2 - \beta_{1n}^2 = 0$" is invalid.*

Proof. Take the additive grid $M + iD + jF$ with $(M, D, F) = (4, 3360, 2112)$: a genuine magic square with magic constant 16428 and center $5476 = 74^2$, six of whose nine entries are perfect squares,

$$\begin{pmatrix} 94^2 & 2^2 & 7588 \\ 4228 & 74^2 & 82^2 \\ 58^2 & 10948 & 46^2 \end{pmatrix}.$$

Its three AP pairs in the preprint's formalism are:

	(p,q,r)	κ	$\alpha=\alpha_n/\alpha_d$	s	N
$\mathcal{P}_1$	$(2,\ 58,\ 82)$	$7/3$	$35/6$	1	16
$\mathcal{P}_2$	$(46,\ 74,\ 94)$	$7/5$	$42/5$	23	4
$\mathcal{P}_3$	$(\sqrt{4228},\ \sqrt{7588},\ \sqrt{10948})$		rep. $\alpha_{3d}:=5$		

Here:

- $\mathcal{P}_1, \mathcal{P}_2$ are *fully integral* AP pairs with common sum $D_1 = 3360$, coprime representations, and all of the preprint's (6)–(8), (11)–(14) satisfied in $\mathbb{Z}$;
- $\mathcal{P}_3$ is *forced by the constraint* (L3.2) *itself*: $p_3^2 = q_2^2 + p_2^2 - q_1^2 = 5476 + 2116 - 3364 = 4228$, $q_3^2 = 7588$, $r_3^2 = 10948$, none a perfect square (as Proposition 1 guarantees must happen for some pair). Exactly, in $\mathbb{Q}(g)$ with $g = \sqrt{105961}$:

$$t^2 = (r_3 - q_3)^2 = 18536 - 56g > 0, \qquad \alpha_3 = \frac{3360}{t^2} = \frac{2317 + 7g}{420},$$

$$s_3^2 = (5\alpha_3 - 15)^2 - 200 = \frac{1057(331+g)}{504} > 0, \qquad N_3 = (5\alpha_3 - 15) - s_3 > 0$$

  (the product of the two N_3-roots is $8\alpha_{3d}^2 = 200$), with $\beta_{2n}^2 = 5\alpha_3/42 > 0$ and $\beta_{2d}^2 = 1$;
- the interleaving is the preprint's own Lemma-3.2 case (c): $\mathcal{P}_2^1 = 92 < 116 = \mathcal{P}_1^2$ and $\mathcal{P}_3^1 = 2\sqrt{4228} < 148 = \mathcal{P}_2^2$, the $\lhd$ relations hold ($2\sqrt{4228} \le 186$, $92 \le 162$), and its case-(c) formulas give $\Sigma\mathbf{p}_A = \Sigma\mathbf{p}_B = 1248$.

The constraint (L3.2) holds by construction — an integer identity, $46^2 - 58^2 = 4228 - 74^2 = -1248$ — so by Theorem A, eq. (29) *holds* at this configuration. By (1) its two sides equal

$$4N_2^2 N_3^2\, \beta_{2n}^2 \beta_{2d}^2\, (\beta_{1d}^2 - \beta_{1n}^2)(\beta_{1n}^2\alpha_{1n}^2 - \beta_{1d}^2\alpha_{1d}^2) = -298392.5892\ldots \neq 0,$$

every factor being nonzero: $N_2^2 = 16$; $N_3^2 \neq 0$; $\beta_{2n}^2\beta_{2d}^2 > 0$; and

$$\beta_{1d}^2 - \beta_{1n}^2 = \frac{5}{6} - \frac{6}{5} = -\frac{11}{30} \neq 0, \qquad \beta_{1n}^2\alpha_{1n}^2 - \beta_{1d}^2\alpha_{1d}^2 = \tfrac{6}{5}\cdot 35^2 - \tfrac{5}{6}\cdot 6^2 = 1440 \neq 0,$$

with $\beta_1 = 24/20 = 6/5$. Meanwhile every positivity cited by the final step ($N_1, N_2, N_3, \beta_{1n}, \beta_{1d}, \beta_{2n}, \beta_{2d} > 0$) holds. The step's hypotheses are satisfied and its conclusion fails. □

Remark. The witness is robust. The representation choice for $\mathcal{P}_3$ is immaterial: taking $\alpha_{3d} = 1$ in place of 5 rescales the $\mathcal{P}_3$-quantities, and (29) again holds with both sides $-477.428\ldots \neq 0$. In a sweep of 500 pseudorandom real grids, (29) held on every one (as Corollary A.1 predicts), with β_1 ranging over $[1.0008,\ 3.998]$ and never equal to 1; and perturbing a single entry of the witness grid by 1 makes (29) fail, confirming empirically that (29) is exactly the constraint.

6. The invalid step, on its own terms

The complete final argument of [1] (p. 7) reads:

> "Compare this with the LHS of equation (29). Clearly, this is a quadratic in α_{1d}^2 and, as such, the coefficients of odd powers of α_{1d} are 0, which must also hold in the RHS. Obviously, $N_1, N_2, N_3, \beta_{1n}, \beta_{1d}, \beta_{2n}, \beta_{2d} > 0$, as per the definitions in equations (11), (21) and (22), so $\beta_{1d}^2 - \beta_{1n}^2 = 0$."

There are exactly two readings, and each fails.

Numeric reading. For any given (hypothetical) magic square, (29) is one equation between two *numbers*: α_{1d} is a single fixed integer, and N_1, N_2, N_3 and the β's are determined by the same data — indeed the preprint's own (11)–(12) define N_1 from α_{1n}, α_{1d}, so the "coefficients" $16\beta_{1n}^2/N_1^2$, $24\beta_{1n}^2/N_1, \ldots$ appearing in its (30) are themselves functions of α_{1d}. Numbers do not have coefficients: the even and odd parts of the two sides of a numerical equality need not separately agree, any more than they do in $3 + 4 = 7$. At the witness of Theorem B, the "odd part" of (30) that the argument declares must vanish evaluates to $-152180.22\ldots$

Polynomial reading. If instead all eight quantities are treated as free variables, so that coefficient comparison would be meaningful, the premise fails: by Theorem A, LHS(29) − RHS(29) is *not* the zero polynomial (it is K_0 times the constraint polynomial), so (29) is not a polynomial identity and there are no coefficients to compare.

There is no third reading: any hypothetical "variation of α_{1d}" drags N_1 and the β's along with it, so the "coefficients" are not well-defined constants under any interpretation.

No local repair is possible. One might hope integrality could rescue the step (the witness's $\mathcal{P}_3$ is irrational). It cannot: (i) the step justifies itself by positivity alone (quoted above), and every positivity it invokes holds at the witness, whose $\alpha_{1d} = 6$ — the comparison variable itself — comes from a fully integral pair; (ii) the preprint itself places $\beta_{1n}, \beta_{1d}, \beta_{2n}, \beta_{2d}$ in $\mathbb{R}^+$, and they enter (29) only squared; (iii) decisively, restricting the step's validity to all-rational configurations is circular: by Proposition 1 the all-rational solution set being empty *is the theorem being proved*, and a step that is sound only if the conclusion is true proves nothing. By Corollary A.1(2), any correct completion of the argument must inject a genuinely new Diophantine ingredient; the preprint contains none (its integrality side conditions — coprime α_n/α_d, perfect-square discriminant — are never used by the derivation, as Theorem A shows).

7. Concluding remarks

This note refutes the proof, not the statement: whether a 3×3 magic square of distinct integer squares exists remains open, in both directions. A byproduct of the analysis seems worth recording: the witness construction generalizes — *any* two integer three-term arithmetic progressions of squares with a common difference D and $q_2^2 + p_2^2 - q_1^2 > 0$ yield, via Proposition 1, a magic square with six perfect-square entries — a clean family of near-misses produced by the preprint's own (correct) framework.

All results above are machine-verified in exact arithmetic — Theorem A by exact integer multivariate polynomial expansion, Theorem B in exact arithmetic over $\mathbb{Q}(\sqrt{105961})$, and the whole analysis independently re-verified through a second implementation following the preprint's definitions verbatim at 60-digit working precision, including the perturbation and random-grid controls of the Remark. Appendix A contains a short self-contained script verifying Theorems A and B; the full verification suite is available from the author on request.

Appendix A. A self-contained verification script

The following script (Python, requiring only the open-source library `sympy`) verifies Theorem A symbolically and Theorem B exactly; it runs in a few seconds.

```
import sympy as sp

n, d, N2, N3, b1n, b1d, b2n, b2d = sp.symbols(
    'n d N2 N3 b1n b1d b2n b2d', positive=True)

# eq. (29), literally as printed (v3, p. 6); b** = beta_**^2:
LHS = N3**2*b2n*b2d*(8*b1d**2*d**2 - N2**2)**2 \
    - N2**2*(8*b2d**2*b1d**2*d**2 - N3**2)**2
RHS = 4*N2**2*N3**2*b1n*b1d*b2n*b2d*(n - d)**2 \
    - 4*N2**2*N3**2*b2n*b2d*(b1n*n - b1d*d)**2

# the Lemma-3.2 constraint 4E/t^2 in the same variables:
a2d, a3d = b1d*d, b2d*b1d*d
T1 = (b2n/b2d) * (8*a2d**2 - N2**2)**2/(4*N2**2) / a2d**2
T2 = (b1n*b2n/(b1d*b2d)) * (n - d)**2 / d**2
T3 = (8*a3d**2 - N3**2)**2/(4*N3**2) / a3d**2
T4 = (b2n/b2d) * (b1n*n - b1d*d)**2 / a2d**2
E4 = T1 - T2 - T3 + T4

# THEOREM A:
assert sp.simplify((LHS - RHS)
                   - 4*N2**2*N3**2*b1d**2*b2d**2*d**2*E4) == 0

# THEOREM B (the witness; exact):
t2 = 18536 - 56*sp.sqrt(105961)
alpha3 = sp.Rational(3360, 1)/t2
vals = {n: 35, d: 6, N2: 4,
        b1n: sp.Rational(6, 5), b1d: sp.Rational(5, 6),
        b2n: 5*alpha3/42, b2d: 1,
        N3: (5*alpha3 - 15) - sp.sqrt((5*alpha3 - 15)**2 - 200)}
lhs_v, rhs_v = LHS.subs(vals), RHS.subs(vals)
assert sp.simplify(lhs_v - rhs_v) == 0   # (29) HOLDS at the witness
assert abs(sp.N(lhs_v, 30)) > 1          # ... both sides NONZERO
print("both sides of (29) =", sp.N(lhs_v, 30),
```

```
    "; beta1^2 = 36/25 != 1")
```

References

[1] O. Hill, *On Arithmetic Progressions and a Proof of the Nonexistence of Magic Squares of Squares*, arXiv:2510.08286v3 (2026).

[2] M. LaBar, Problem 270, *College Mathematics Journal* **15** (1984), 69.

[3] R. K. Guy, *Unsolved Problems in Number Theory*, 2nd ed., Springer, 1994, Problem D15.